Lab 4 — Clustering, PCA and the ML I Project

Machine Learning I · Tutorial

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Wednesday, September 30, 2026 · 19:00–22:00

Goal of the session. Implement k-means and watch its objective fall exactly as Lecture 6's proof says it must; verify Ward's merge formula on real clusters; compute principal components two ways and confirm they agree; check the variance–reconstruction identity numerically; and close ML I with the project.

Materials

- `lab_04_student.ipynb` / `lab_04_solution.ipynb`.
- Sources: CS 155 Problem Set 5 (PCA, SVD, low-rank approximation).
- Data: `make_blobs`, `load_digits` (1797×64), `load_diabetes` — all bundled.

1 k-Means (40 min)

Task 1 — Lloyd's algorithm, and the lecture example

Write `kmeans(X, mu0, iters)` that alternates the two half-steps of Lecture 6 and records J after each. Run it on the six points $A(1,1)$, $B(1.5,2)$, $C(3,3.5)$, $D(5,7)$, $E(3.5,5)$, $F(4.5,5)$ from $\boldsymbol{\mu}_1 = A$, $\boldsymbol{\mu}_2 = D$. You must get $J = 22$ after the first assignment and $J = 9.167$ at convergence, with final centroids $(1.833, 2.167)$ and $(4.333, 5.667)$. Assert that the recorded J sequence is non-increasing.

Task 2 — local minima, and k-means++

On `make_blobs(300, centers=4, cluster_std=0.9)`, run your `kmeans` from 50 random initializations (centroids drawn from the data) and histogram the final J. How many distinct local minima do you find? Then implement k-means++ seeding (the D^2 sampling rule) and repeat. Compare the two histograms and the best J from each. Finally confirm `sklearn.cluster.KMeans` with your best seeds reaches the same J.

Task 3 — choosing K

For $K = 1, \ldots, 10$ on the same data, plot J (elbow) and the mean silhouette (`sklearn.metrics.silhouette_score`). Do they agree on $K = 4$? Then add one far outlier to the data and repeat. What happens to the elbow, and why (Lecture 6, "What k-Means

Assumes")?

2 Hierarchical Clustering (25 min)

Task 4 — Ward's formula, verified

Take two clusters A and B from Task 2's data (any two of the four). Compute the within-cluster sum of squares J with them separate and with them merged, and check that the increase equals $\frac{n_A n_B}{n_A+n_B}\left\|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\right\|^2$ to floating-point precision. Then compute `scipy.cluster.hierarchy.linkage(X, "ward")` and confirm its last merge height, squared, equals the ΔJ of merging the final two clusters (scipy reports $\sqrt{2\Delta J}$; check its documentation and reconcile the factor).

Task 5 — four linkages, one dendrogram each

Draw the dendrograms for single, complete, average and Ward linkage on 40 points from three blobs. Cut each at three clusters and compare the partitions to the truth with the adjusted Rand index. Which linkage fails, and on what kind of data would it be the right choice?

3 PCA (50 min)

Task 6 — two computations, one answer

Centre the digits data $\mathbf{X}$ (1797×64).

1. Eigendecompose $\mathbf{S} = \mathbf{X}^\top\mathbf{X}/N$ with `np.linalg.eigh`; sort descending.
2. Compute the thin SVD of $\mathbf{X}$; verify $\mathbf{V}$ matches the eigenvectors (up to sign) and $\lambda_j = d_j^2/N$.
3. Verify the Ky Fan claim: for $k = 5$, $\operatorname{tr}(\mathbf{V}_k^\top\mathbf{S}\mathbf{V}_k) = \lambda_1 + \cdots + \lambda_5$, and it exceeds $\operatorname{tr}(\mathbf{W}^\top\mathbf{S}\mathbf{W})$ for 100 random orthonormal $\mathbf{W} \in \mathbb{R}^{64\times 5}$ (orthonormalize with `np.linalg.qr`).

Task 7 — variance = total − reconstruction

For $k = 1, 2, 4, 8, 16, 32, 64$ compute (i) the mean squared reconstruction error $\frac{1}{N}\sum_i \left\|\mathbf{x}_i - \mathbf{V}_k\mathbf{V}_k^\top\mathbf{x}_i\right\|^2$, (ii) $\operatorname{tr}\mathbf{S} - \sum_{j\le k}\lambda_j$. They must agree. Plot the proportion of variance explained against k; report the smallest k reaching 90%.

Task 8 — eigendigits

Reshape the first eight principal components to 8×8 and display them. Reconstruct one digit with $k = 1, 2, 4, 8, 16, 32, 64$ components and display the sequence. At which k can you read the digit?

Task 9 — hard vs. soft: PCR against ridge

On the diabetes data (standardized), compare by 10-fold CV: principal-components regression with $k = 1, \ldots, 10$ components, and ridge with λ on a log grid. Plot both CV curves against effective degrees of freedom (k for PCR; $\sum_j d_j^2/(d_j^2+\lambda)$ for ridge). Lecture 6 called PCR a hard threshold and ridge a soft one; which wins here, and by how much?

4 The ML I Project (due today, 23:59)

One dataset, the whole pipeline, a short report. Choose one:

- `load_breast_cancer` — binary classification, 569×30.
- `load_digits` — 10-class classification, 1797×64.
- `load_diabetes` — regression, 442×10.
- The Messidor retinopathy set from CS 155 Set 3, at `reference/caltech_cs155/problem_sets/set3/messidor_features.arff`.

Deliverable

A notebook and a PDF report of at most 4 pages, containing:

1. Setup — the task, the loss you are optimizing, the metric you will report, and why they differ if they do. The test split, sealed before anything else.
2. Baseline — the trivial predictor, and its number.
3. Three models — one regularized linear model (ridge/lasso/logistic), one random forest, one gradient-boosting model, each tuned by cross-validation with the search grid stated. Standardization inside folds.
4. Evaluation — the chosen metric on the test set for all four, once. For classification, also a confusion matrix and ROC/AUC; for regression, residual plots.
5. One derivation — pick any result from Lectures 1–6 that your pipeline relies on (the optimism formula, the bagging variance, the soft-threshold rule, …) and re-derive it in your own words in half a page.
6. What you would do next — three sentences.

Rubric (out of 20): setup and protocol 4, baseline and models 6, evaluation 4, derivation 4, clarity 2. A pipeline that touches the test set more than once loses the 4 protocol points, whatever its numbers.

5 Exercises (optional; not graded)

Exercise 1. For standardized 2-D data with correlation $\rho > 0$, show that

$$\mathbf{S} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

has eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$ with eigenvalues $1+\rho$ and $1-\rho$. Verify numerically.

Exercise 2. Implement power iteration for the top eigenvector of $\mathbf{S}$: $\mathbf{u} \leftarrow \mathbf{S}\mathbf{u}/\|\mathbf{S}\mathbf{u}\|$. Show that the angle to `eigh`'s answer decays like $(\lambda_2/\lambda_1)^t$ on the digits data.

Submission

Notebook `lab04_lastname_firstname.ipynb`, and the project as `project_lastname_firstname.ipynb` + `.pdf`, to the course drive before Wednesday, September 30, 23:59.